

Anshuman Pandey

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PROFESSIONAL SUMMARY

CSE (AI/ML) student at VIT Bhopal with hands-on experience building deep learning and computer vision systems. Developed production-grade ML models achieving up to 98.2% accuracy. Proficient in Python, TensorFlow, and OpenCV. Actively seeking internship or entry-level roles in Machine Learning Engineering or Data Science.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — Vellore Institute of Technology, Bhopal

2023 – Present | CGPA: 8.57/10

Class XII: 89.5% | Class X: 89.4%

PROJECTS

AlphaZ0 – Neural Chess Engine | Python, PyTorch, MCTS, A3C, Pygame | 2026

- Built a fully playable chess engine from scratch with complete rule support (castling, en passant, promotion, legal move filtering) and an 18-plane board encoder for neural input.
- Designed a ResNet dual-head neural network (10 residual blocks, 128 channels) with separate policy and value heads, modeled after DeepMind's AlphaZero architecture.
- Implemented MCTS with UCB selection and Dirichlet noise at root, combined with a parallel A3C training loop where worker threads compute GAE-advantage gradients asynchronously.
- Built a Pygame UI with main menu, algebraic notation move log sidebar, bot-vs-human mode with color selection, promotion dialog, and board flip for black.

<https://github.com/AnshumanJ28/AlphaZ0>

Diabetic Retinopathy Detection | Python, PyTorch, EfficientNet-B5, timm | 2025

- Fine-tuned a pretrained EfficientNet-B5 on the APTOS 2019 and RFMiD retinal fundus image datasets for binary classification of diabetic retinopathy risk.
- Applied two-phase transfer learning — classifier-head-only training followed by full fine-tuning at reduced LR — using AdamW with cosine annealing scheduling.
- Handled class imbalance via weighted BCE loss and augmentation (flips, rotation, colour jitter); evaluated with accuracy and MCC for robust performance on imbalanced medical data.

<https://github.com/AnshumanJ28/Diabetic-B5>

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL (MySQL), HTML, CSS

ML / AI: TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, CNN, LSTM, FFT preprocessing

Tools & Platforms: AWS, Git, Jupyter Notebook, Advanced Excel, Canva, N8N

Concepts: Data Structures & Algorithms, Computer Vision, Signal Processing, Front-end Development

CERTIFICATIONS

C & C++ Programming (Udemy, HackerRank) | **Python & Java** (HackerRank) | **Cloud Computing** (NPTEL) | **Advanced Excel** (Udemy) | **AI** (TalentSprint) | **ML** (Coursera) | **AWS trainer**(AWS)

ACHIEVEMENTS

- 2nd Runner-up, IEEE-NSUT Ideathon — awarded for ML-based project among 80+ competing teams.
- Top 10 finish, BITS Pilani Hackathon — among top teams nationally in a competitive problem-solving event.

EXTRA-CURRICULAR ACTIVITIES

Team AVITAR Member | AARAMBH Chess Competitor | Robotics Workshop Participant